

CogAge all-subject optimizer comparison

Classes and availability

Bending=0, Walking=1, Sitting=2, Standing=3. Four-class optimizer runs are limited to subjects with paired training and testing folders for all four classes. In this dataset those are S1, S2, S5, and S7. S3, S6, and S8 are included in inspection but excluded from four-class experiments because selected activity pairs are missing. S4 has no original test folders.

Data and splits

Each subject dataset stores 128 x 33 windows. Original _1_extracted folders supply train; _2_extracted folders supply held-out test. Validation Events are selected only from each subject's original train windows, with Event groups kept intact. A new StandardScaler is fitted only on the model-training portion and applied to validation/test. No class balancing is used; class counts are shown below.

Network and fixed settings

NumPy MLP: 4224 -> 32 -> 16 -> 4 logits, ReLU hidden activations, 135,796 parameters, stable sparse cross-entropy from logits. Initialization seed 42; 100 epochs; learning rate 0.01; mini-batch 8; momentum 0.9. RMSprop uses decay 0.99, Adam/AdamW use betas 0.9/0.999, AdamW weight decay 0.01.

Optimizer update ideas

Full/mini-batch/stochastic GD use batch-mean gradients with the selected update frequency. Momentum accumulates a velocity; Adagrad accumulates squared gradients; RMSprop uses an exponential average of squared gradients; Adam combines corrected first and second moments; AdamW applies decoupled weight decay.

Cleaning and limitations

The per-subject raw audit records malformed rows, missing values, timestamp intervals, interpolation, synchronized row removals, and discarded incomplete windows. Each subject is a separate participant, within subject results are not participant-independent. Few Events make Validation Scores noisy. Test Scores are descriptive and must not be used for hyperparameter tuning.

Subject	Optimizer	Val loss	Epoch	Test acc.	Macro F1	Seconds	Updates
S1	Full-batch GD	1.001	100	0.850	0.842	4.00	100
S1	Mini-batch GD	0.999	100	0.850	0.842	9.35	400
S1	Stochastic GD	0.999	100	0.850	0.842	7.87	3,200
S1	SGD + Momentum	1.545	1	0.825	0.791	8.70	400
S1	Adagrad	0.615	2	0.900	0.899	6.46	400
S1	RMSprop	4.477	1	0.800	0.799	6.46	400
S1	Adam	0.764	4	0.925	0.925	7.59	400
S1	AdamW	0.756	4	0.925	0.925	8.16	400
S2	Full-batch GD	2.106	3	0.750	0.683	5.15	100
S2	Mini-batch GD	1.859	4	0.800	0.757	11.84	400
S2	Stochastic GD	0.502	3	0.975	0.975	7.59	3,200
S2	SGD + Momentum	1.282	3	0.825	0.825	22.87	400
S2	Adagrad	0.043	1	0.950	0.948	9.08	400
S2	RMSprop	3.454	5	0.900	0.896	14.78	400
S2	Adam	2.242	1	0.900	0.893	7.86	400
S2	AdamW	2.240	1	0.900	0.893	9.00	400
S5	Full-batch GD	0.618	85	0.800	0.750	2.86	100
S5	Mini-batch GD	0.176	74	0.775	0.705	5.30	400
S5	Stochastic GD	0.571	6	0.875	0.875	5.85	3,200
S5	SGD + Momentum	0.000	10	0.750	0.661	4.74	400
S5	Adagrad	0.054	100	0.850	0.849	4.79	400
S5	RMSprop	0.000	2	0.625	0.577	7.68	400
S5	Adam	0.006	4	0.750	0.723	10.64	400
S5	AdamW	0.006	4	0.750	0.723	10.85	400
S7	Full-batch GD	0.138	100	0.615	0.540	4.10	100
S7	Mini-batch GD	0.057	100	0.667	0.588	15.02	400
S7	Stochastic GD	0.000	100	0.744	0.661	7.08	3,000
S7	SGD + Momentum	0.063	1	0.641	0.572	5.99	400
S7	Adagrad	0.000	100	0.718	0.642	7.41	400
S7	RMSprop	-0.000	1	0.744	0.696	11.30	400
S7	Adam	0.000	3	0.744	0.661	8.36	400
S7	AdamW	0.000	3	0.744	0.661	7.66	400

Validation loss curves

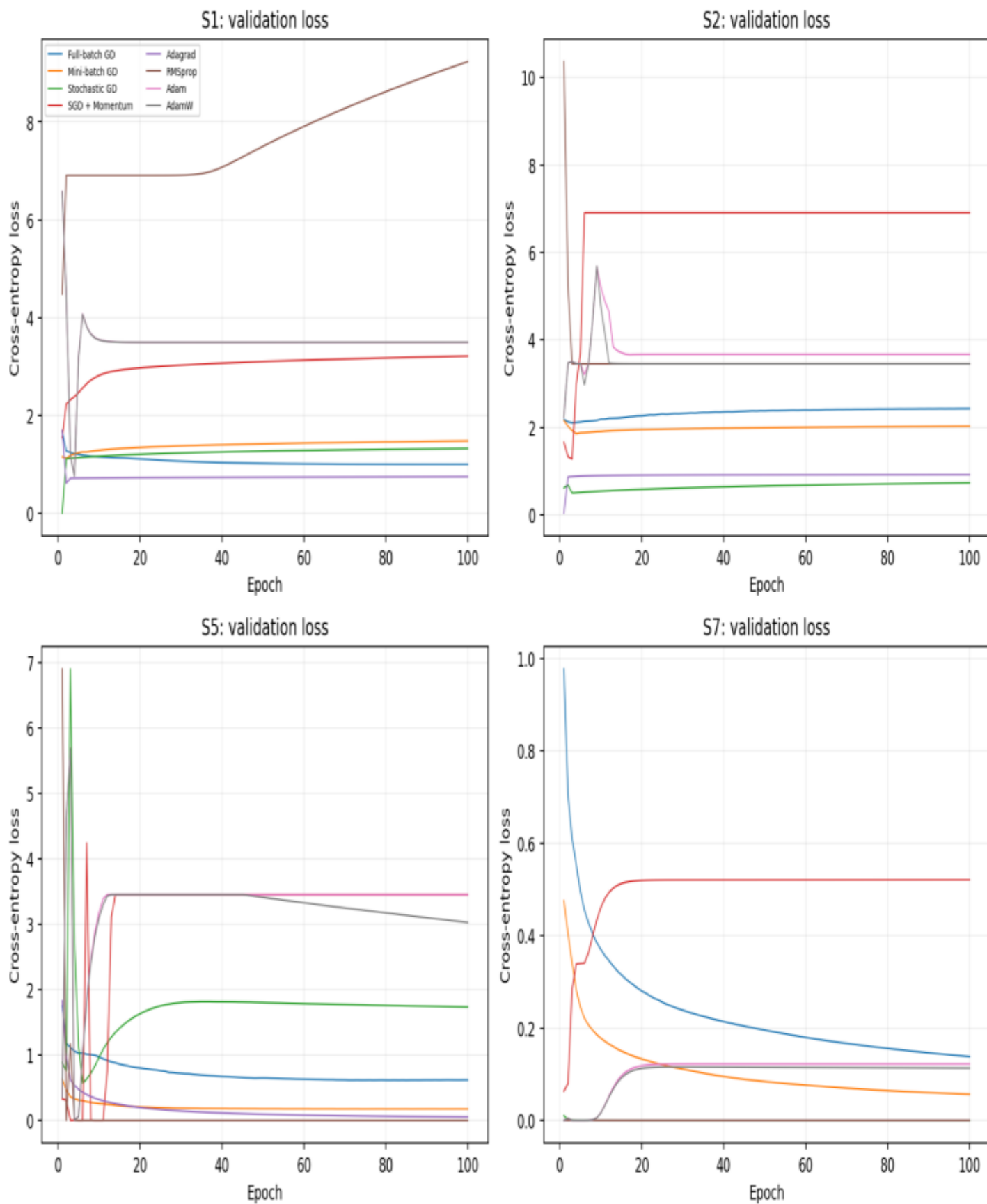


Figure 1. Validation loss by epoch for the four complete subjects. Each panel shows the eight optimizer runs.

Validation accuracy curves

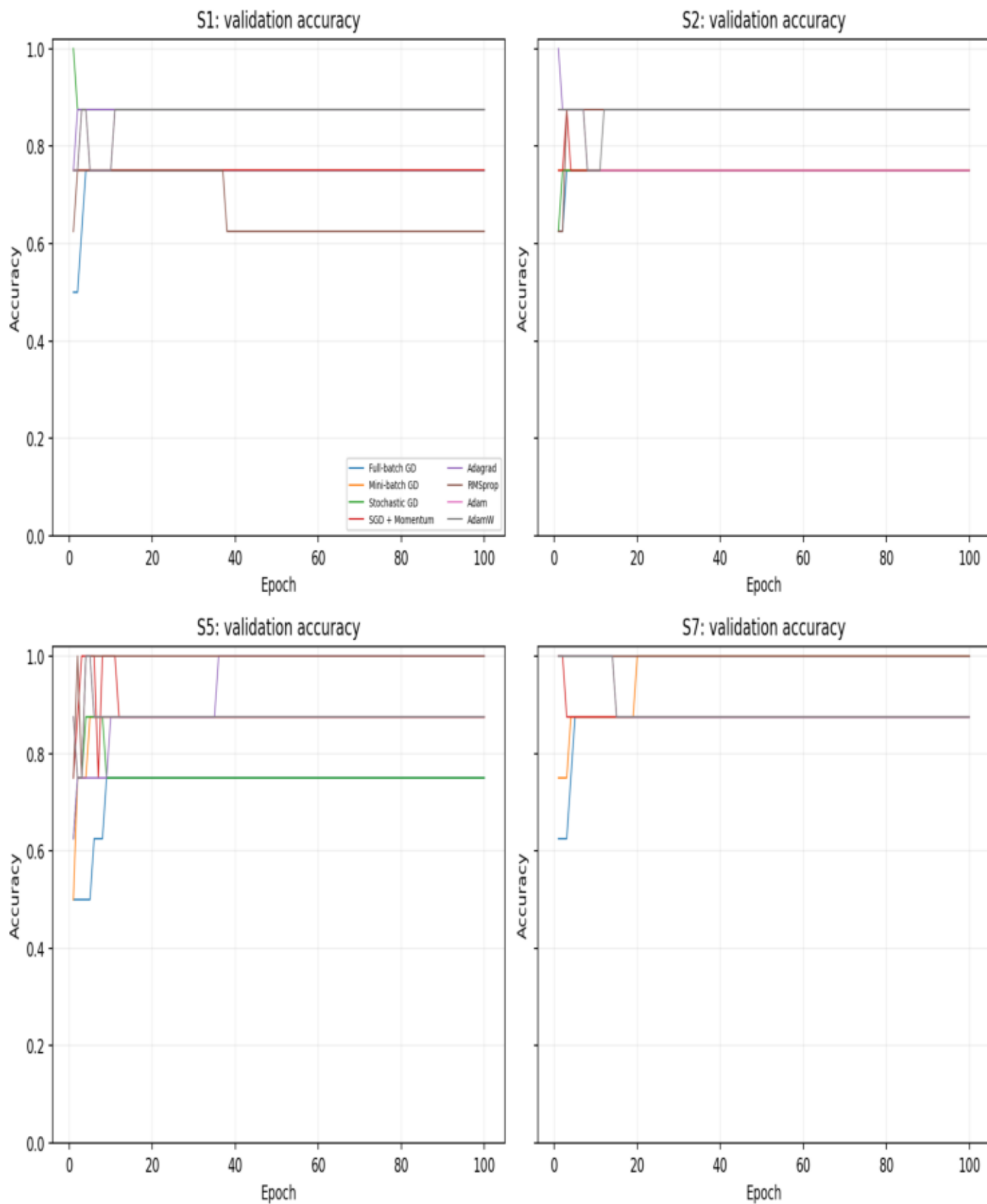


Figure 2. Validation accuracy by epoch. Validation sets are small; one incorrect window can noticeably change accuracy.

Convergence and held-out test metrics

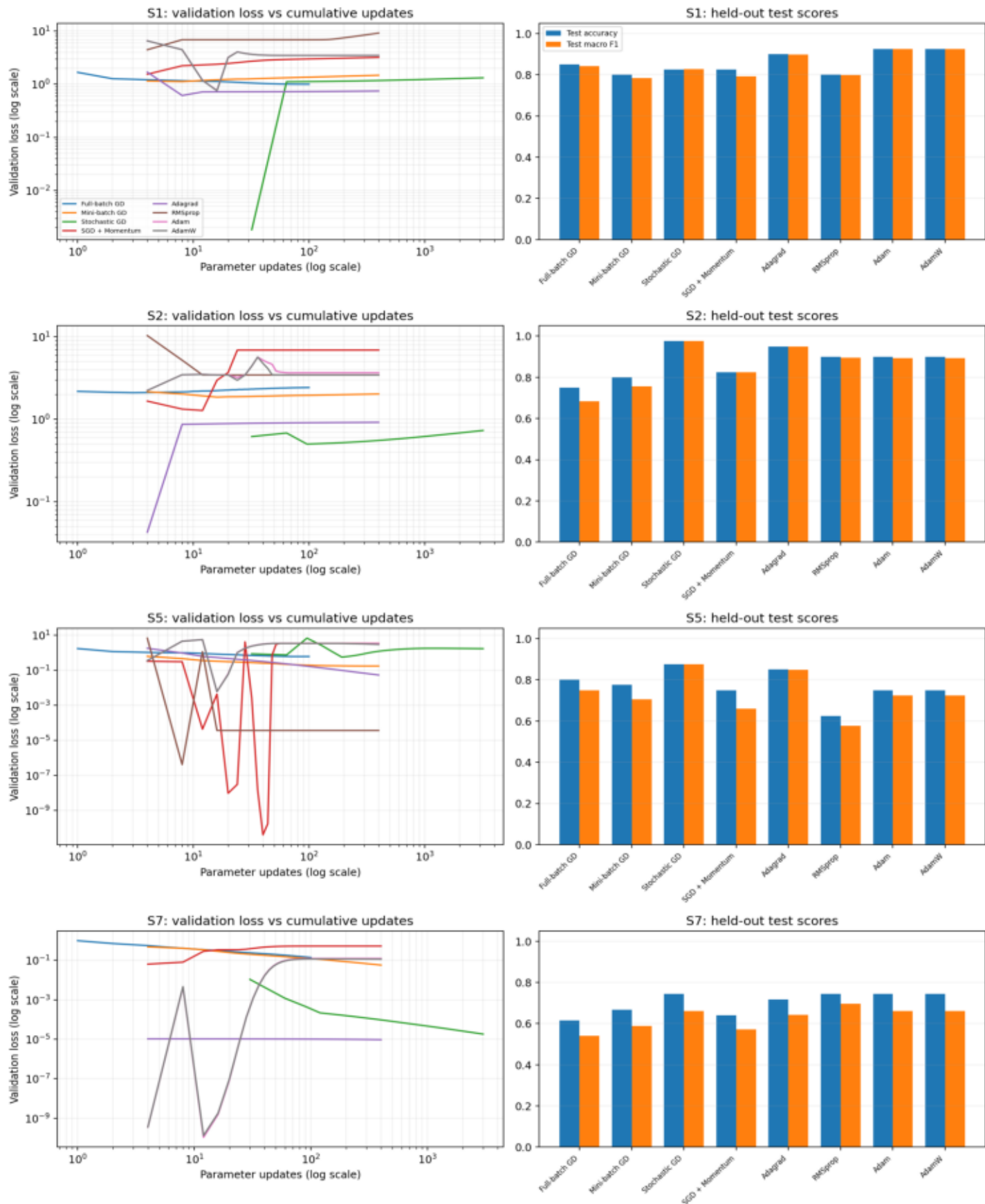


Figure 3. Validation loss versus cumulative updates (left) and held-out test accuracy/macro F1 (right), per subject. Update counts differ with batch size.

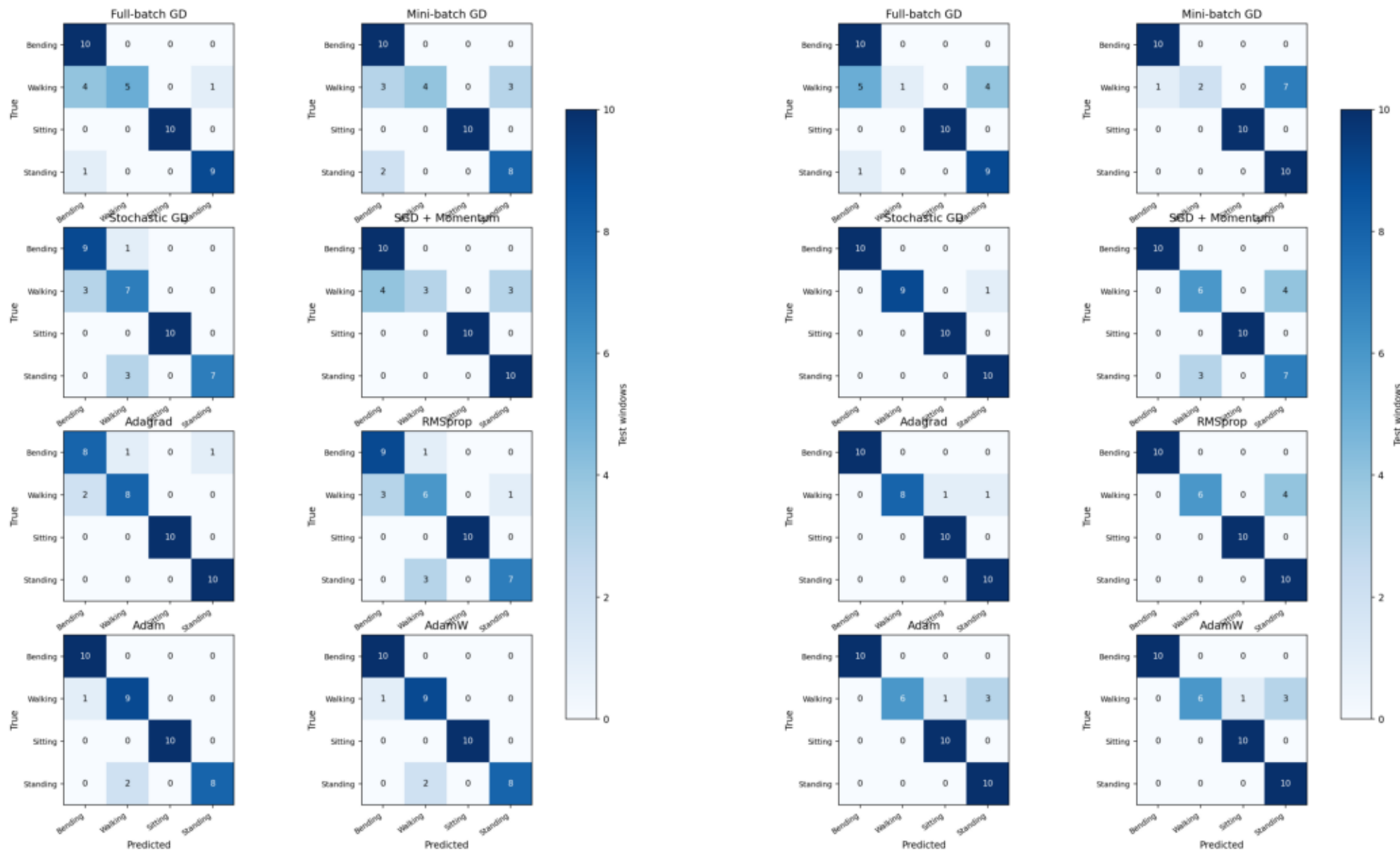
Held-out test confusion matrices: S1, S2

S1

S2

S1: held-out test confusion matrices (rows=true, columns=predicted)

S2: held-out test confusion matrices (rows=true, columns=predicted)



Each matrix is 4 x 4; row=true class, column=predicted class; label order: Bending, Walking, Sitting, Standing.

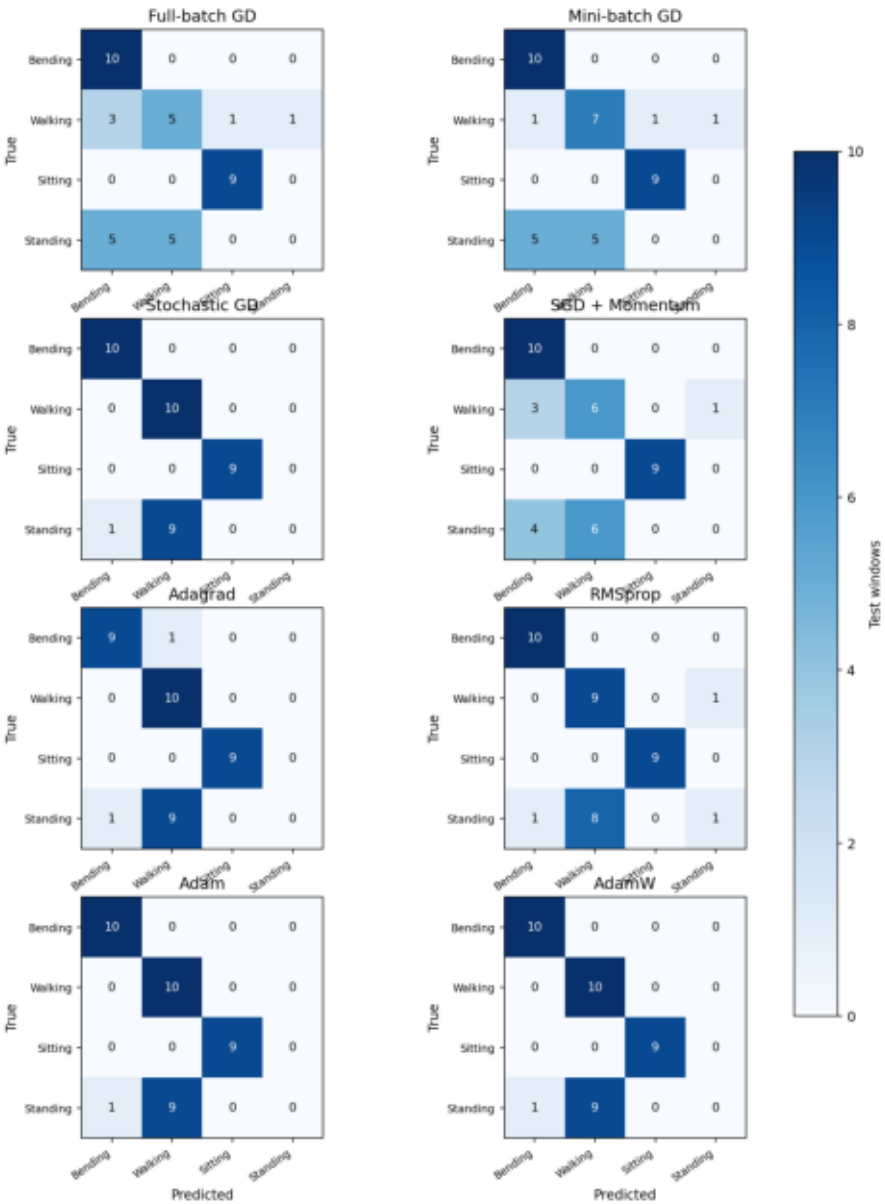
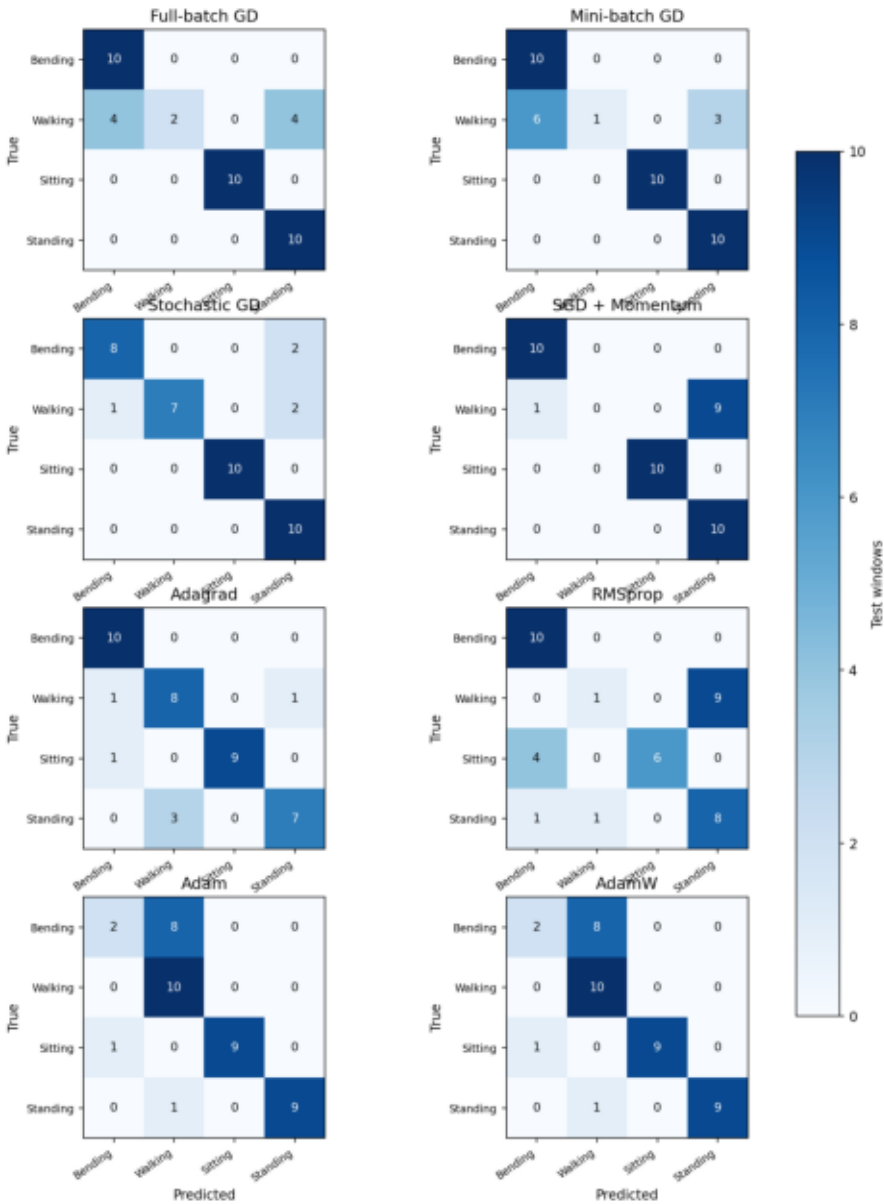
Held-out test confusion matrices: S5, S7

S5

S7

S5: held-out test confusion matrices (rows=true, columns=predicted)

S7: held-out test confusion matrices (rows=true, columns=predicted)



Each matrix is 4 x 4; row=true class, column=predicted class; label order: Bending, Walking, Sitting, Standing.

Interpretation and reproducibility

Observed outcomes

S1: lowest validation loss was Stochastic GD (0.002, epoch 1); highest test macro F1 was Adam (0.925). | S2: lowest validation loss was Adagrad (0.043, epoch 1); highest test macro F1 was Stochastic GD (0.975). | S5: lowest validation loss was SGD + Momentum (0.000, epoch 10); highest test macro F1 was Stochastic GD (0.875). | S7: lowest validation loss was RMSprop (-0.000, epoch 1); highest test macro F1 was RMSprop (0.696).

Fair comparison

Full-batch GD performs one update per epoch, mini-batch runs perform $\text{ceil}(N/8)$, and stochastic GD performs N . Equal epoch counts therefore mean different update counts. Use the saved update-based plots and elapsed-time table; do not claim speed from epoch curves alone.

Reading optimizer differences

Adaptive methods scale updates using gradient history; Momentum adds directional persistence. AdamW differs from Adam by applying weight decay separately from the adaptive gradient update. The dataset is too small for a broad optimizer ranking; interpret per-subject differences cautiously.

Artifacts and execution

Notebook execution writes CSV histories, summary metrics, JSON confusion matrices, per-subject model weights and plots to `run_20261003_103105`. Run `prepare_all_subjects.py` first, then execute all notebook cells from top to bottom. Random seed is 42.

AI assistance disclosure

GitHub Copilot assisted with drafting and debugging code. Review the pipeline and report, and disclose this assistance as required by course policy.